

1. An Autonomous Robotic System for Object Retrieval and Delivery (MDPI Robotics, 2026)

Link: <https://doi.org/10.3390/robotics15020041>

Overview:

This paper presents a mobile manipulator built on a Pioneer P3-DX base fitted with a ReactorX-200 robotic arm, targeted at assisting people with disabilities and older adults who cannot physically retrieve items themselves. The system accepts a spoken request (e.g. "bring me my pen"), maps the requested object to a semantic location (kitchen, office, bedroom), autonomously navigates there using a 270° LiDAR and the ROS move_base navigation stack, performs a local exploration routine (moving and rotating in place) to visually locate the object using YOLOv8, estimates its position with a Kinect depth camera, repositions itself if the object is beyond the arm's 550 mm reach, and finally executes a grasp-and-deliver sequence.

Methodology/Platform: Pioneer P3-DX mobile base + ReactorX-200 arm, YOLOv8 object detector, AprilTag-assisted localization, Kinect depth sensing, ROS navigation stack.

Project Objective:

To design and validate an integrated mobile-manipulator system that lets a user verbally request a household object and have the robot autonomously find, grasp, and deliver it, thereby increasing independent living for disabled and elderly users.

Research Gap:

1. Prior assistive robots typically solved navigation, object detection, and manipulation as isolated sub-problems rather than as one coordinated pipeline.
2. Voice-based interaction had rarely been tightly coupled with semantic object-to-room mapping for retrieval tasks.
3. Existing systems lacked a robust method for handling the case where an object is detected but lies outside the arm's physical reach, requiring dynamic repositioning.

Research Questions:

1. How can voice-based human robot interaction be leveraged to enable intuitive and accessible communication for object retrieval tasks in home environments?
2. Which robot navigation algorithm is most effective for enabling reliable autonomous mobility in cluttered and dynamic household settings?
3. How can deep learning-based object detection and AprilTag-assisted localization provide robot perception for identifying and retrieving user-requested items?
4. How can an integrated mobile manipulator combining a Pioneer P3-DX base and a ReactorX-200 robotic arm be coordinated to achieve grasping and delivery of objects to the user?

Key Contributions:

1. A semantic keyword-to-location mapping table (object → room) that narrows the robot's search space before physical exploration begins.
2. Integration of YOLOv8 detection with AprilTag localization for precise end-effector-frame coordinate transformation.
3. An autonomous "reach-check-and-reposition" behavior that bridges navigation mode and grasping mode.
4. End-to-end empirical validation of the full request → navigate → detect → grasp → deliver pipeline on physical hardware.

Limitations noted: Reliance on pre-tagged semantic object-location mappings means novel/unmapped objects may not be found efficiently; performance in highly cluttered or visually ambiguous scenes is not deeply stress-tested.

2. CamFi: An AI-driven and Camera-based System for Assisting Users in Finding Lost Objects in Multi-Person Scenarios (ACM CHI EA, 2022)

Link: <https://dl.acm.org/doi/10.1145/3491101.3519780>

Overview:

CamFi is a fixed (non-mobile) camera-based AI monitoring system designed for shared spaces (offices, shared apartments) where multiple people's belongings intermingle. Rather than deploying a mobile robot, it passively watches a room and logs when/where personal items were last observed, letting users query "where did I last see my keys?" It builds on a lineage of RFID- and vision-based item-finding research (e.g. IteMinder, Ferret RFID localization) but focuses specifically on the multi-user ownership-disambiguation problem.

Project Objective:

To reduce the time people spend searching for misplaced items in multi-occupant environments by using passive visual monitoring instead of requiring every object to be individually tagged (as RFID-based systems do).

Research Gap:

1. Earlier lost-and-found aids (RFID tags, single-user object trackers such as Ferret/IteMinder) do not scale well to spaces shared by multiple people with visually similar belongings.
2. Camera-based systems had not adequately addressed the ownership-attribution problem - i.e., determining whose item is which when several similar objects are present.

3. Most prior work assumed a single user's home rather than a shared/communal environment (offices, dorms, libraries).

Research Questions:

1. How can a camera-based system disambiguate ownership of visually similar objects among multiple users sharing a space?
2. How accurately can such a system recall an object's last-seen location and time without relying on dedicated per-object tags?
3. What interface design best supports users querying "last seen" information in a multi-person context?

Key Contributions:

1. A continuous video-logging architecture that records object appearance events without requiring RFID tagging.
2. A "last-seen" retrieval interface tailored to multi-person, multi-object scenarios.
3. Groundwork for occupancy-aware object search that a mobile lost-and-found robot could later adopt to avoid confusing different users' belongings.

Limitations noted: Being a fixed-camera system (not a mobile robot), coverage is limited to camera field-of-view; blind spots and occlusion remain unresolved.

3. Finding Misplaced Items Using a Mobile Robot in a Smart Home Environment (*Frontiers of Information Technology & Electronic Engineering*, 2019)

Link: <https://link.springer.com/article/10.1631/FITEE.1800275>

Overview:

This paper introduces a "Misplaced Item Finding" (MIF) system that combines smart-home sensing with mobile robotics. Instead of having the robot search the entire house exhaustively, it fuses the resident's historical trajectory data (captured via multi-sensor fusion in a smart-home testbed) with a **CNN-based real-time object detector** to intelligently prioritize the areas most likely to contain the lost item - i.e., places the person actually visited recently.

Methodology: Multi-sensor fusion for resident localization/tracking → trajectory-informed path planning → CNN-based object detection at candidate locations. Validated experimentally in a smart-home testbed.

Project Objective:

To use knowledge of where a person has recently moved through the home to constrain and

speed up a robot's search for a misplaced object, rather than relying on brute-force room-by-room scanning.

Research Gap:

1. Smart-home sensing (occupancy/trajectory tracking) and mobile-robot vision-based object detection had largely been studied as separate research threads.
2. No prior system had combined human movement history with real-time CNN detection specifically to accelerate misplaced-item search.
3. Existing object-search robots lacked a principled way to rank candidate search locations by probability derived from human behavior data.

Research Questions:

1. Can human historical trajectory data meaningfully constrain and shorten a robot's physical search path for a misplaced item?
2. Does a CNN-based object detector reliably confirm the presence of the misplaced item once the robot reaches a candidate area?
3. How efficient and accurate is the combined trajectory + detection approach compared to undirected search, as measured in a real smart-home testbed?

Key Contributions:

1. A multi-sensor fusion method for localizing and tracking a resident within a smart home.
2. A path-planning algorithm that incorporates historical trajectory knowledge to generate an efficient robot search route.
3. A real-time CNN-based object detector tuned for confirming misplaced items at candidate locations.
4. Empirical validation in a smart-home testbed demonstrating measurable gains in search accuracy and efficiency over undirected search.

Limitations noted: Dependent on the smart home already having trajectory-sensing infrastructure installed; less useful in homes without such sensor coverage.

4. CleanNav: Deep Learning and Reinforcement Strategies for Smart Robot Exploration (*ScienceDirect*, 2026)

Link: <https://www.sciencedirect.com/science/article/pii/S2590005626000457>

Overview:

CleanNav repurposes an autonomous vacuum-cleaning robot platform, which already traverses most of a home routinely, into a passive "lost & found" logging system. As it cleans, it uses an

RL-optimized navigation/exploration model alongside object detection to log every object it observes - including label, timestamp, confidence score, and local context - so a user can later query the last-known location of a specific item. It also uses a robotic arm to physically move obstructing items during cleaning.

Methodology: Hierarchical architecture - ultrasonic sensors for immediate obstacle avoidance, an RL model for navigation/exploration optimization, and a robotic-arm + object-detection module for physical interaction with detected items.

Project Objective:

To leverage a cleaning robot's incidental full-home coverage as a byproduct data source for object tracking, benefiting users in scenarios where small items habitually go missing.

Research Gap:

1. Household service robots (e.g. vacuum robots) already achieve near-total home coverage during normal operation, but this incidental coverage was not being exploited as an object-location logging service.
2. Reinforcement-learning-based exploration strategies had not been combined with an object-logging subsystem in commercial-style cleaning robots.
3. No prior system captured contextual metadata (confidence scores, local context, timestamps) at the density needed to answer "where did I last see X" queries reliably.

Research Questions:

1. Can a cleaning robot's routine exploration double as a passive object-location logging service without disrupting its primary cleaning task?
2. How should confidence scores, timestamps, and local context be structured and stored to make "last known location" queries useful and reliable?
3. How does an RL-based navigation/exploration strategy compare to conventional navigation in terms of both cleaning efficiency and object-logging completeness?

Key Contributions:

1. A hierarchical robustness architecture separating immediate obstacle avoidance, RL-optimized exploration, and object-interaction tasks.
2. A real-time object-logging system recording labels, timestamps, confidence scores, and contextual/local information.
3. Demonstration that a single-purpose service robot (vacuum) can be extended into a dual-purpose lost-and-found utility with minimal added hardware.

Limitations noted: Detection is opportunistic (only logs what it happens to see during cleaning passes), so very recently misplaced items may not yet be logged, log-based approach doesn't perform active search on demand.

5. "Where Is My Phone?" — Towards Developing an Episodic Memory Model for Companion Robots to Track Users' Salient Objects (ACM/IEEE HRI, 2023)

Link: <https://doi.org/10.1145/3568294.3580160>

Overview:

Developed at the University of Waterloo's Social and Intelligent Robotics Research Lab, this paper builds an episodic memory system on a Fetch mobile manipulator, targeted at people with dementia and memory impairment. As the robot moves through the home performing its normal functions, it detects and tracks salient personal objects (phone, glasses, medication) via camera, logging when items enter/leave its field of view. A companion smartphone/computer app lets a caregiver or user query an object by name and receive its last-known time and location.

Methodology: Fetch robot + camera-based object detection/tracking tuned to discriminate salient items from visual distractors and clutter; graphical interface for object queries.

Project Objective:

To build a companion robot with its own artificial episodic memory that can help people with cognitive decline locate important everyday objects, reducing burden on caregivers.

Research Gap:

1. Assistive robots for people with cognitive decline had mostly focused on navigation, mobility assistance, or social companionship - not on building a searchable memory of the *objects* the user tends to lose.
2. Object-detection systems struggled with distinguishing salient, user-relevant items from surrounding clutter and distractor objects.
3. No prior work had proposed an "episodic memory" model specifically structured for long-horizon object-location recall in a home setting.

Research Questions:

1. How can a robot robustly detect and track salient objects amid visual clutter and moving distractor objects?
2. How should an episodic memory model be structured to support reliable, long-horizon object-location recall?
3. How usable and trustworthy is the resulting system for its target population (dementia patients and caregivers)?

Key Contributions:

1. An object-detection and tracking pipeline specifically tuned for salient-object discrimination (rather than generic object detection).

2. A graphical/app-based query interface for retrieving an item's last-seen time and place.
3. High reported detection/recall accuracy in initial testing, with a stated roadmap toward user studies with people without disabilities, then people with dementia.

Limitations noted: The robot (quite bulky, per the authors) still needs miniaturization for real household deployment; testing with the actual target population (dementia patients) had not yet been conducted at time of publication.

6. Autonomous Robot Retrieval System (Ahern, Carter & Wilson, 2015)

Link:

https://www.researchgate.net/publication/314095433_Autonomous_Robot_Retrieval_System

Overview:

This project builds a small, low-cost mobile robot (Lego NXT platform) that performs autonomous object retrieval in a completely unknown environment, combining **OpenRatSLAM** (a biologically inspired SLAM algorithm) with vision based object recognition. It is explicitly framed as a counterpoint to the trend of large, expensive SLAM capable robot platforms.

Methodology: RatSLAM-based simultaneous localization and mapping on Lego NXT hardware, paired with a vision-based object recognizer, performance benchmarked using a MATLAB RatSLAM implementation.

Project Objective:

To demonstrate that inexpensive, off-the-shelf hardware can still perform SLAM-driven autonomous search and object retrieval in unfamiliar environments.

Research Gap:

1. Most SLAM-capable robotic platforms at the time were large, sophisticated, and costly, limiting accessibility for smaller-scale or educational projects.
2. There was limited evidence on whether biologically inspired SLAM (RatSLAM) could run effectively on constrained, low-cost embedded hardware.
3. Integration of lightweight SLAM with vision-based object recognition on such minimal hardware had not been thoroughly benchmarked.

Research Questions:

1. Can a lightweight, biologically inspired SLAM implementation (RatSLAM) run effectively on low-cost hardware for object-search tasks?

2. How well does vision-based object recognition integrate with this SLAM pipeline in a completely unknown environment?
3. What are the real-time performance costs (e.g. processing time relative to video duration) of running such an implementation on constrained hardware?

Key Contributions:

1. A low-cost mobile robot platform combining SLAM mapping with object recognition for retrieval tasks.
2. Empirical timing analysis showing the MATLAB RatSLAM implementation ran roughly 10x slower than the video duration it processed - a useful negative/constraint finding for real-time system designers.
3. A practical demonstration that resource-limited hardware can still achieve autonomous retrieval, albeit with performance trade-offs.

Limitations noted: Runtime performance (10x slower than real-time) makes the specific implementation impractical for real-time deployment without further optimization or hardware upgrades.

7. Efficient Dynamic Object Search in Home Environment by Mobile Robot: A Priori Knowledge-Based Approach

Publication details: Y. Zhang, G. Tian, J. Lu, M. Zhang, S. Zhang - *IEEE Transactions on Vehicular Technology*, Vol. 68, No. 10, pp. 9466–9477, 2019.

Link: <https://ieeexplore.ieee.org/document/8794623/>

Overview:

This is a full journal-length paper (not a short conference paper), which explains its depth. It tackles a very specific, practical robotics problem: a mobile robot needs to find a dynamic object (one that gets moved around, unlike furniture) somewhere in an entire house, and doing so by visiting every room and scanning exhaustively is prohibitively slow and energy-costly. The authors' insight is explicitly anthropomorphic: humans don't search randomly — if you lose your phone, you first check the couch, then the kitchen counter, then your bag — because you have implicit "common sense" about where things of that type tend to end up. The paper formalizes this common sense as a priori knowledge.

Technical Architecture:

1. **A priori knowledge base:** Encodes the statistical/likely spatial relationship between an object category and rooms/other objects (e.g. "keys are commonly near the door or on tables").
2. **Cost-aware room prioritization:** Rather than only ranking rooms by probability of containing the object, the algorithm explicitly computes a trade-off between inferred location probability and the physical distance/cost of traveling to that candidate room meaning a moderately-likely-but-nearby room can be preferred over a highly-likely-but-far room if the expected search cost favors it.
3. **In-room search heuristics:** Once inside a candidate room, the search space is *further reduced* using a priori knowledge-guided heuristics (rather than scanning the whole room uniformly), narrowing down to likely sub-locations (e.g. near furniture known to be associated with that object type).
4. **Knowledge update mechanism:** The paper explicitly investigates how the a priori knowledge base itself should be updated over time to maintain reliability i.e. the system is designed to adapt as it learns that its assumptions were wrong (e.g. if an object is repeatedly found somewhere unexpected).
5. **Validation:** Implemented and evaluated in both simulation and real-world environments with "extensive experiments" cited to demonstrate feasibility and efficiency gains.

Project Objective:

To design a search strategy that lets a mobile robot locate a dynamic (frequently relocated) target object across an entire home at minimum total cost (time/distance), by exploiting common-sense spatial priors the way a human searcher would.

Research Gap:

1. Object search is a prerequisite capability for household robots, but the dynamic nature of objects (they move) combined with the scale and complexity of home environments made low-cost search an unsolved problem prior to this work.
2. Existing search methods did not jointly optimize for *both* inferred location probability *and* physical travel distance, they typically optimized one or the other, leading to inefficient real-world search paths.
3. There was no established mechanism for maintaining/updating a priori knowledge over time as a robot's experience revealed inaccuracies in its assumptions.

Research Questions:

1. How can common-sense spatial knowledge about object-room and object-object relationships be formally modeled as reusable a priori knowledge for a robot's search planning?
2. How should a search algorithm balance the probability that an object is in a given room against the travel cost of reaching that room, to minimize total expected search cost?
3. Once inside a candidate room, how can a priori knowledge further reduce the in-room search space rather than requiring exhaustive scanning?

4. How can the reliability of the knowledge base be maintained/updated as the robot accumulates real-world search experience?

Key Contributions:

1. A formalized a priori knowledge model of typical object-location and object-object relationships, directly inspired by human search behavior.
2. A novel cost-aware room-prioritization search strategy - the key technical novelty distinguishing this from prior methods that ignored travel distance.
3. A secondary in-room heuristic search mechanism to reduce search space *within* a chosen room.
4. A knowledge-update mechanism that keeps the system's assumptions accurate over long-term deployment.
5. Empirical validation in both simulated and real physical environments - a stronger evidence bar than simulation-only papers.

Relevance to a Lost & Found Assistant Robot:

This paper is arguably the single most directly transferable IEEE paper for your project: it is precisely the "active search once an item is reported missing" module your system would need, and its cost/probability trade-off logic could directly inform how the robot decides which room to check first.

Noted Limitation:

Performance depends on the quality of the a priori knowledge base; atypical households (unusual furniture arrangement, unconventional storage habits) may reduce the model's accuracy until it has adapted through the knowledge-update mechanism.

8. Hierarchical Semantic Knowledge-Based Object Search Method for Household Robots

Publication details: M. Zhang, G. Tian, Y. Cui, Y. Zhang, Z. Xia - *IEEE Transactions on Emerging Topics in Computational Intelligence*, Vol. 8, No. 1, pp. 930–941, 2024.

Link: <https://ieeexplore.ieee.org/abstract/document/10196338>

Overview:

This is a 2024 journal follow-on to the research lineage above - same broader research group context (Guohui Tian et al.), extending the a priori knowledge idea into a more structured, hierarchical semantic framework. Where Paper 1 modeled a relatively flat probability/cost trade-off, this paper explicitly builds a three-level relationship: (**room category** → **environmental object** → **dynamic object**). The idea: humans don't just think "the phone is probably in the bedroom" - they reason "the phone is probably on the nightstand, and the nightstand is in the bedroom." This is a deeper, more human-like chain of association.

Technical Architecture:

1. **Semantic knowledge construction:** Built by modeling the (room category, environmental object, dynamic object) relationship, extracted from **natural language** descriptions relevant to home services (e.g. text corpora or common-sense linguistic resources describing typical household layouts).
2. **Ambiguity mitigation layer:** Natural language is inherently ambiguous/lossy (e.g., "near the table" could mean several different tables). To handle this, the authors construct a secondary a priori knowledge layer specifically linking room categories to environmental objects, disambiguating what the semantic/linguistic layer alone cannot resolve.
3. **Search guidance mechanism:** The target object is associated with "relevant but obvious" objects or room categories - meaning the robot doesn't need to have ever seen the exact target object before; it can reason about where it's likely to be based on nearby, easily-recognizable "anchor" objects (e.g. recognizing a nightstand is much easier and more reliable than recognizing a specific phone, and the nightstand tells you where to look).
4. **Evaluation:** The authors report experimental results demonstrating "feasibility and effectiveness" of applying this knowledge-based framework to service-planning-oriented object search.

Project Objective:

To reduce the inefficiency of object search caused by home-environment complexity and dynamics, by building a semantic reasoning hierarchy that lets a robot infer a target's likely location even when the object cannot be directly observed at first.

Research Gap:

1. Object search remains a basic but unsolved necessity for household robots because complex, dynamic home environments often prevent target objects from being observed immediately, directly reducing search efficiency.
2. Prior work lacked a hierarchical structure connecting room categories, fixed environmental objects, and dynamic objects - meaning reasoning could not chain the way human reasoning does (object → likely furniture → likely room).

3. Natural-language-derived semantic knowledge inherently introduces ambiguity and information loss, and no established method addressed this specifically in the object-search context.
4. Relying purely on direct recognition of the target object is fragile; there was a gap in methods that let a robot infer location via easier-to-recognize related objects instead.

Research Questions:

1. How can a three-level hierarchical semantic relationship (room category, environmental object, and dynamic object) be constructed and represented in a form usable by a robot's search planner?
2. How can the ambiguity and information loss inherent in natural-language-derived relationships be resolved using a supplementary a priori knowledge layer?
3. Does search guided by "anchor object" association (finding the nightstand to infer the phone's likely location) outperform methods that rely on direct target-object recognition or flat (non-hierarchical) priors?

Key Contributions:

1. Construction of a hierarchical semantic knowledge base explicitly modeling (room category, environmental object, dynamic object) relationships derived from natural language.
2. A disambiguation layer of a priori knowledge that resolves the ambiguity/information loss inherent to natural-language semantic sources.
3. A generalized search-guidance method that reduces dependence on directly recognizing the (potentially small, occluded, or novel) target object itself, by leveraging easier-to-detect related anchor objects and room types.
4. Experimental demonstration of feasibility/effectiveness for service-planning-oriented household object search.

Relevance to a Lost & Found Assistant Robot:

This paper is highly relevant for handling cases where your robot's camera can't directly see or recognize the exact lost item (e.g. small items, items in odd orientations, poor lighting) - it shows how to still narrow down the search using bigger, easier-to-recognize furniture/room cues instead.

Noted Limitation:

Building and maintaining an accurate semantic hierarchy for diverse home layouts, cultures, and habits likely requires either significant manual curation of the natural-language knowledge source or a robust learning pipeline - a practical deployment cost not deeply addressed in the abstract-level description.

9. Object Search Using Edge-AI Based Mobile Robot

Publication details: Ryosuke Miyata, Osamu Fukuda, Nobuhiko Yamaguchi, Hiroshi Okumura - presented at the *International Conference on Intelligent Informatics and BioMedical Sciences (ICIIBMS) 2021* (IEEE-published proceedings); also appears in the *Journal of Information and Communication Engineering (JICE)*, Vol. 7, No. 2, pp. 455-461, Dec. 2021.

Link: <https://ieeexplore.ieee.org/document/9651591/>

Overview:

Unlike the two knowledge-based papers above (which are about *where* to search), this paper is about how the robot is built and controlled - specifically, running the entire object-search pipeline on-device using an edge-AI computer rather than depending on cloud servers. The system is entirely open-source and ROS-based, and supports two coupled capabilities: **voice recognition** (to receive a spoken instruction naming the object) and **object recognition** (to detect and approach that named object in the environment).

Technical Architecture:

1. **Hardware/software base:** Built using the **Robot Operating System (ROS)** to facilitate development - an open architecture choice that improves reproducibility and lowers the barrier for other researchers/teams to build on the work.
2. **Onboard edge-AI computer:** All core functions - voice recognition, object recognition, and motor control - are implemented directly on this edge computer, avoiding reliance on internet connectivity or cloud inference servers.
3. **Voice-driven object approach:** The robot is controlled via spoken commands; once the user names an object, the system uses its object-recognition module to detect and physically approach it.
4. **Evaluation method:** The authors ran preliminary experiments specifically to verify the performance of the developed system (implying quantitative or qualitative performance checks on real hardware, though the abstract-level source does not detail exact metrics).

Project Objective:

To develop and validate an autonomous mobile robot that can perform voice-commanded object search and approach entirely through onboard edge-AI processing, eliminating cloud dependency and improving responsiveness for real-time household use.

Research Gap:

1. Cloud-dependent robotic AI pipelines introduce network latency and connectivity dependency, both of which are poor fits for real-time, in-home object search tasks.

2. There was limited prior work demonstrating that voice recognition, object recognition, *and* motor control could all run simultaneously and reliably on constrained onboard (edge) hardware, rather than being distributed across cloud services.
3. Fully open-source implementations combining these three subsystems were relatively rare, limiting reproducibility and making it harder for the field to build on such systems.

Research Questions:

1. Can an edge-AI computer handle voice recognition, object recognition, and motor control concurrently with acceptable real-time performance, without cloud offloading?
2. Does an open-source, ROS-based architecture generalize well across different objects and different voice commands?
3. What is the practical, real-world performance of such a system when deployed on physical hardware (as opposed to simulation)?

Key Contributions:

1. A fully open-source, ROS-based robot architecture integrating voice recognition, object recognition, and motor control entirely on an onboard edge-AI computer.
2. Demonstrated feasibility of intuitive voice-command-driven object search without any cloud dependency - directly relevant to privacy-sensitive or connectivity-limited household deployments.
3. Preliminary real-hardware experimental verification, rather than purely simulated results.

Relevance to a Lost & Found Assistant Robot:

This paper maps closely onto the interaction and hardware-architecture layer of your project: it demonstrates a practical, low-cost, ROS-based blueprint for the "user says the object name → robot goes and finds it's interaction loop, without needing cloud infrastructure - useful if the assistant robot needs to work offline or with minimal latency.

Noted Limitation:

The authors themselves describe the experiments as "preliminary" implying limited scale/diversity of test objects, environments, and voice-command variety, the abstract does not report accuracy/latency benchmarks, so its performance ceiling relative to cloud-based systems is unclear.

10. Cognitive Learning Enabled Real Time Object Search Robot

Publication details: C. Sadhu, M. H. Abhiram, B. Chandan, et al. - presented at the *International Conference on Control, Automation, Robotics and Embedded Systems (CARE) 2013*, pp. 1-6.

Link: <https://doi.org/10.1109/CARE.2013.6733739>

Overview:

This is the earliest paper in the set (2013) and represents a foundational, short conference contribution rather than a full journal study. It's frequently cited by later, more detailed papers (via the Springer FITEE review), which is a strong signal that it is regarded as an early conceptual precursor in this specific research lineage - i.e., the idea of applying cognitive/learning-based reasoning (rather than brute-force or purely reactive vision-based scanning) to real-time object search predates, and likely influenced, the more sophisticated priori-knowledge and semantic-hierarchy approaches developed later.

Technical Architecture:

Given this is a short 2013 conference paper with limited detailed public abstract text available beyond citation records, the technical specifics are less granular than for Papers 1-3. What can be reliably established:

1. The system is explicitly framed as "cognitive learning enabled" - i.e. incorporating a learning component into the search-decision process, rather than static, hand-coded search rules.
2. It targets real-time operation, implying design constraints around computational efficiency typical of embedded robotics of that era (2013 hardware).
3. It is cited alongside RFID-based (IteMinder), planning-based (occluded target search), and other object-search approaches in survey-style literature reviews on misplaced/object-search robotics, positioning it as a distinct "learning-based" branch within that broader taxonomy.

Project Objective:

To demonstrate that a cognitive-learning approach - where the robot's search behavior can improve through learning rather than relying purely on exhaustive or reactive scanning - can meaningfully reduce real-time object-search time.

Research Gap:

1. At the time of publication, object-search robots largely relied on exhaustive scanning or purely reactive vision-based methods, without any mechanism for the robot to learn and improve its search strategy over repeated tasks.

2. There was no established framework demonstrating that a learning-based reasoning component could operate within the tight computational/time budget required for **real-time** object search on embedded robotic hardware.
3. The broader field lacked empirical groundwork connecting cognitive-learning theory to practical, real-time robotic object-search deployment - a gap that later papers (semantic/priori-knowledge based) built upon.

Research Questions:

1. How can cognitive/learning-based reasoning reduce the time required for real-time object search, compared to exhaustive or purely reactive scanning methods?
2. What learning architecture allows a robot to incrementally improve its search performance across repeated search tasks?
3. Can such a cognitive-learning approach satisfy the real-time computational constraints required for practical embedded robot deployment?

Key Contributions:

1. One of the earliest documented real-time, cognitive-learning-based frameworks specifically applied to object-search robotics.
2. Established conceptual and empirical feasibility of applying learning-based reasoning under real-time constraints on embedded robotic systems - a foundation that is directly extended in the later priori-knowledge (2019) and hierarchical-semantic (2024) papers reviewed above.
3. Contributed to the broader taxonomy of object-search strategies (alongside RFID-based and planning-based methods) referenced in subsequent survey literature.

Relevance to a Lost & Found Assistant Robot:

Best used in your review as the **historical/foundational anchor** - showing that "search via learning rather than brute force" is not a new idea in 2019 or 2024, but has over a decade of research lineage, strengthening the literature review's narrative of a clear, evolving research trajectory (2013 → 2019 → 2024).

Noted Limitation:

As an early-stage, it is a 6-page conference paper, its evaluation scale, environmental diversity and technical detail are inherently more limited than the journal-length studies. It helped inspire, detailed methodology and quantitative results are not as fully accessible from public abstract-level sources.

